

ROVER LAB, DOSSIER (TRAPPIST-1 g mission)

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Planet

Name: TRAPPIST-1 g

Planet type: Cold rocky planet at the outer edge of the TRAPPIST-1 habitable zone

Surface temperature: ≈ -76 °C equilibrium estimate. Tidally locked.

Surface pressure: Not measured; no atmosphere detected

Gravity: ≈ 1.04 g (essentially Earth-like)

Atmosphere: Unconfirmed. Theoretical models suggest either a global ice-covered ocean or a thick primordial steam atmosphere, but neither has been observed.

Mission Briefing

TRAPPIST-1 g is the coldest of the temperate worlds in the TRAPPIST-1 system, likely locked in a permanent deep freeze except for a narrow band near its sunward face. The rover 'Terra-g', a sister unit to the one exploring TRAPPIST-1 e closer to the star, has reached a pale, faintly tinted ice-crust flat near the terminator, where a thin, briny slush seeps up through cracks in the ice. Given how little evaporation happens on such a cold world, mission control expects any dissolved salts here to be far more dilute than on TRAPPIST-1 e, but even a weak signal of iron would be worth confirming for a future outpost.

Rover

Starting energy: 100

Site visit cost: 10

Experiment cost: 15

Overview Image

Site: Ice-crust flat with briny seepage

Description: A pale, frost-covered flat cut by thin cracks where a faintly tinted slush seeps to the surface.

Identity: Iron(III) chloride solution (real sample, suggested ≈ 0.05 M), the most dilute of the five real groups. REAL SAMPLE GROUP.

Known Properties

Colour: Very pale yellow

Odour: None detectable

Solubility: Fully miscible with water

Experiment Results

Measure pH: Weakly acidic, pH around 3.0–3.3

Measure density: 1.02 g/mL, only slightly denser than water

Test electrical conductivity: Low to moderate

Add a magnet: No attraction

Add potassium thiocyanate solution: Turns a pale pink-red

Add sodium hydroxide solution: Forms a faint, fine reddish-brown precipitate

Heat a small sample gently: No visible change

Experiments

Measure pH

Measure density

Test electrical conductivity

Add a magnet

Add potassium thiocyanate solution

Add sodium hydroxide solution

Heat a small sample gently

AI Behaviour

Reveal remaining candidates after narrowing: No

Socratic questions: Yes

Difficulty: Standard (15-16 years old)

Teacher Notes

REAL SAMPLE GROUP, assign real FeCl₃, the most dilute of the five (suggested ≈ 0.05 M). TRAPPIST-1g pairs directly with TRAPPIST-1e (same star system, different orbital distance), a nice opportunity for those two groups to compare notes at the final conference: same star, same likely iron-rich core hypothesis, but very different surface conditions and (by design) very different brine concentrations and iron oxide yields.

Real-science notes: TRAPPIST-1g is a genuine rocky exoplanet in the same system as TRAPPIST-1e, discovered in the same 2017 study (Spitzer + TRAPPIST telescope), orbiting every 12.35 days at ~ 40.7 light-years from Earth. Its density (5.04 g/cm³) is close to Earth's, consistent with (like the rest of the system) an unusually large iron core for its size (Agol et al. 2021). It sits at the outer/cooler edge of the optimistic habitable zone. No atmosphere has been detected; published 'global ocean' or 'steam atmosphere' scenarios for this planet are theoretical models only, not observations.

Sources: Agol et al. 2021 (see en.wikipedia.org/wiki/TRAPPIST-1g for citations); NASA Exoplanet Archive (exoplanetarchive.ipac.caltech.edu/overview/TRAPPIST-1).

Cross-subject link: Biology selective pressure for this planet, permanent extreme cold, possibly beneath a global layer of ice.